CSE 2025
Designer Genes

By: Evergreen Valley High School

Section 1	/24
Section 2	/27
Section 3	/45
Combined score	/96

Section 1: Multiple Choice & Short Answer Section

Answer the following questions to the best of your ability. There is exactly 1 correct answer for each question, and each question is worth 1 point in this section.

1) The ABO blood group is a classification used to categorize individuals based on antigens present on red blood cells into three groups: A, B, and O. Blood types are determined by the combination of 2 alleles, with alleles I^A (blood type A) and I^B (blood type B) being codominant with each other and completely dominant over i (blood type O). A mother with "A" blood type has two children with a father of AB blood type. What is the probability both of the children have blood type B?
    a) 0
    b) ¼
    c) ½
    d) ¾
    e) 1

2) Duchenne muscular dystrophy (DMD) is a genetic disease characterized by progressive muscle weakness and muscle degeneration and follows a X-linked recessive pattern of inheritance. A mother with DMD has a boy with an unaffected father. What is the probability the child is affected by DMD?
    a) 0
    b) ¼
    c) ½
    d) ¾
    e) 1

3) A bacterial strain has undergone several mutations and is now incapable of performing replication. The genetic material of this bacteria shows correct base pairings for DNA, but some parts are fragmented. Which enzyme is likely no longer functioning in the cell?
    a) DNA Polymerase I
    b) DNA Polymerase II
    c) DNA Polymerase III
    d) DNA gyrase
    e) DNA ligase

4) Blue petals (B) are dominant to yellow eyes (b). A blue-petalled plant is crossed with a yellow-petalled plant and produces only blue petalled plants. What is the phenotypic ratio of the F2 generation?
    a) 0% blue-petalled plants; 100% yellow petalled plants
    b) 25% blue-petalled plants; 75% yellow petalled plants
    c) 50% blue-petalled plants; 50% yellow petalled plants
    d) 75% blue-petalled plants; 25% yellow petalled plants
    e) 100% blue-petalled plants; 0% yellow petalled plants

5) What type of mutation is likely to be the most harmful to a eukaryotic cell?
    a) Nonsense mutation 3 amino acids before the stop codon
    b) Missense mutation which changes the codon to code for leucine instead of isoleucine
    c) Deletion of 9 nucleotides in an intron
    d) Silent mutation at the start codon
    e) Frameshift mutation after 5 codons

6) Neurofibromatosis type 1 (NF1) is a condition caused by the mutation to the NF1 gene. Two heterozygous individuals affected by NF1 are capable of producing both affected and unaffected offspring. If at least one of the parents is homozygous dominant for NF1, the child is guaranteed to be affected by NF1. Individuals affected by NF1 experience can develop symptoms, including cafe au lait spots, Lisch nodules (tiny bumps under the eyes), neurofibromas, and short stature. Which of the following correctly describes the mutation to the NF1 gene?
    a) Incomplete dominance
    b) Polygeny
    c) Pleiotropy
    d) Anticipation
    e) Variable Expressivity

7) Which of the following is an accurate description of degeneracy in the context of genetics?
    a) Mutations can cause the gradual collapse of a species due to inactivation of important genes.
    b) Unedited mRNA can cause proteins to misfold due to additional amino acids.
    c) Cellular conditions allow for the same codon to code for more than one amino acid.
    d) Ribosomes are incapable of translating certain sequences of codons, preventing the synthesis of certain proteins.
    e) Multiple codons can specify a specific amino acid through wobble base pairing.

8) Why is rRNA used for phylogenetic analysis compared to other RNA molecules?
    a) Because of its conserved stem-loop structures and sequence variability, rRNA can be used to achieve higher resolution in phylogenetic trees.
    b) rRNA in species evolve under relaxed constraint, and an organism's molecular clock can be used to track evolutionary history.
    c) rRNA is scarce in prokaryotic cells, allowing for scientists to ignore the Bacteria and Archaea domains.
    d) Translational selection does not affect rRNA transcripts because it is a non-coding region, creating a conserved region for comparison.
    e) rRNA genes are located in chromosomes whose synteny (localization in the chromosome) is conserved across species.

9) What correctly explains the difference between cis- and trans-acting factors in gene expression?
    a) Cis-acting factors can only affect a single allele. Trans-acting factors are able to affect both alleles.
    b) Cis-acting factors are typically proteins which affect gene expression. Trans-acting factors are typically RNA molecules that bind to DNA.
    c) Cis-acting factors bind to and modify ribosomes. Trans-acting factors bind to and modify histone tails.
    d) Cis-acting factors are defined by their mobility. Trans-acting factors are defined by their fixed position.
    e) Cis-acting factors are the translated parts of mRNA. Trans-acting factors are the untranslated parts of mRNA.

10) What is the role of the SRY gene in human sex determination?
    a) It is involved in the development of Mullerian ducts.
    b) It represses the development of testis when 2 X chromosomes are present.
    c) It promotes the synthesis of estrogen for male embryos.
    d) It repairs the sex chromosomes during replication to ensure a healthy phenotype.
    e) It is involved in initiating the development of testis by creating a certain transcription factor.

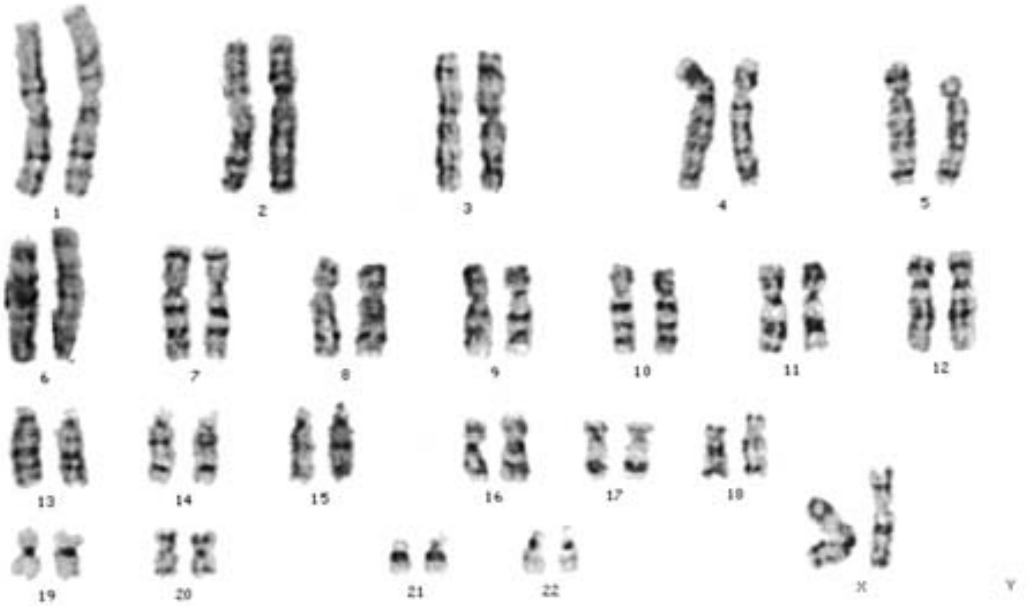
11) Androgens are a group of sex hormones which are produced more abundantly in males than females. Complete androgen insensitivity syndrome (CAIS) is a condition where androgen receptors in males are defective and cannot bind ligands properly. Which of the following correctly matches an individual's genotype with their phenotype, assuming they are affected by CAIS?
    a) Genotype: XX; Phenotype: Male
    b) Genotype: XX; Phenotype: Intersex
    c) Genotype: XX; Phenotype: Female
    d) Genotype: XY; Phenotype: Female
    e) Genotype: XY; Phenotype: Male

12) Diploids carry 2 sets of chromosomes (2n). What is the first point in meiosis in which cells become monoploid and carry a single cell set of chromosomes (n)?
    a) At the end of anaphase I
    b) At the end of prophase II
    c) At the end of telophase I
    d) At the end of prophase I
    e) At the end of interkinesis

13) A mother with genotype AABbCcDDEe has a child with a father with genotype aabbCCDDEe. Assume all genes are independent and complete dominance for all alleles. What is the likelihood the child exhibits only dominant phenotypes?
    a) 0
    b) 0.125
    c) 0.250
    d) 0.375
    e) 0.500

14) A type of flower has 2 genes required for a teal color. If either gene A or B is homozygous recessive, the flower will be white. What is the expected phenotypic ratio of dihybrid cross between the teal flowers (teal:white)?
    a) 3:1
    b) 11:5
    c) 9:7
    d) 13:3
    e) 10:6

15) According to the karyotype below, what disease does this person have?

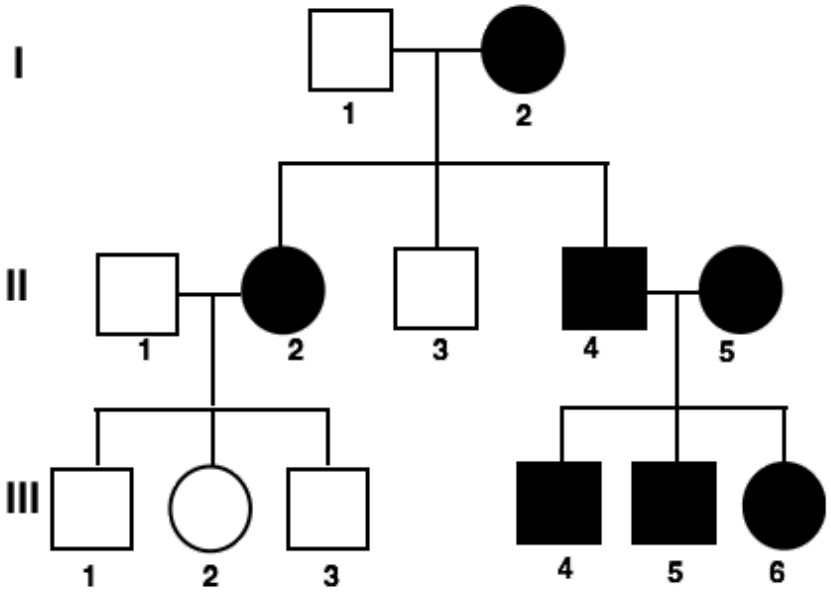


a) Kleinfelter Syndrome
b) Down Syndrome
c) Cri du Chat Syndrome
d) Patau Syndrome
e) None of the above

16) What properly explains why people affected with Turner syndrome are typically infertile?

a) The lack of a Barr body results in the overexpression of X-linked genes.
b) The X chromosome becomes inactivated and the ovaries never develop.
c) Meiosis is halted in the individual due to the insufficient number of X chromosomes, preventing any eggs from forming.
d) Streak gonads form as a result of an absent second X chromosome in place of ovaries.
e) Sex cells fail to migrate to the genital ridge due to the deletion of the SRY gene.

Use the pedigree tree to answer the following questions.



17) According to the pedigree tree, what is the <u>most likely</u> mode of inheritance?
   a) Autosomal dominant
   b) Autosomal recessive
   c) X-linked dominant
   d) X-linked recessive
   e) Y-linked (holandric)

18) In Sanger sequencing, what specifically allows DNA elongation to terminate?
   a) Use of dideoxynucleotides which lack a 3' hydroxyl group
   b) Use of restriction enzymes to degrade DNA strands into sequenceable units
   c) Addition of special primers that attract enzymes which inhibit elongation
   d) Use of heat to denature enzymes to prevent DNA elongation
   e) Use of ribonucleotides to create unstable sections of DNA

19) Your friend is trying to engineer a plasmid that contains genes coding for a fluorescent protein in *E. coli*. Which of the following choices is a requirement for the engineered plasmid to have in order to properly be transcribed and translated?
   a) A lac promoter to recruit RNA polymerase for transcription
   b) TATA box in order for RNA polymerase to bind upstream of the genes of interest
   c) A Shine-Dalgarno sequence in the DNA of the plasmid for proper translation
   d) An oriT site to initiate proper transcription
   e) Inclusion of intron sequences to separate genes

20) Which of the following is true about chloroplast DNA (cpDNA)?
    a) cpDNA is typically linear like eukaryotic chromosomes.
    b) cpDNA is organized into nucleosomes.
    c) cpDNA is maternally inherited.
    d) cpDNA contains mostly inactivated genes as a result of evolution.
    e) cpDNA is transcribed by nuclear RNA polymerase.

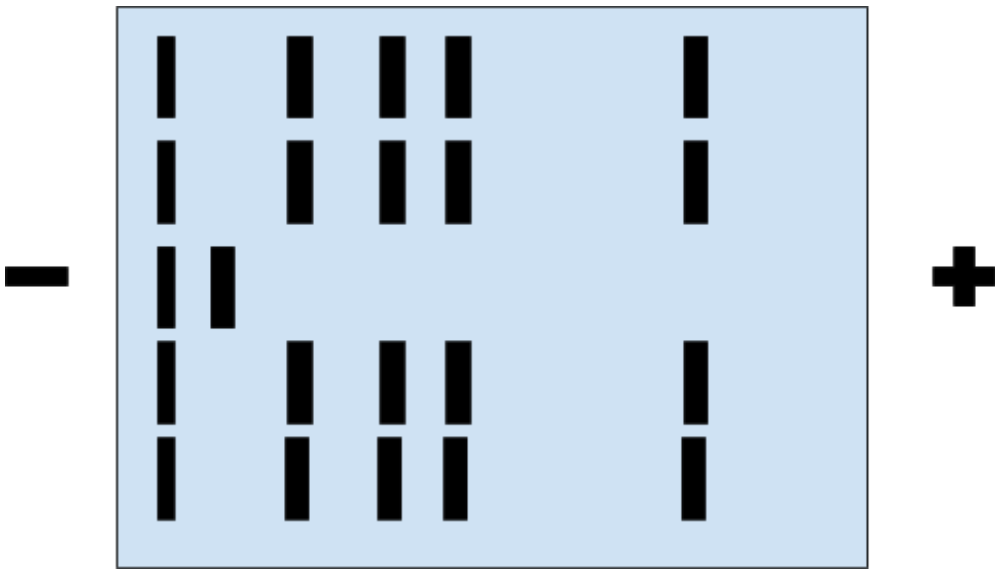
21) You're researching mitochondrial DNA (mtDNA) and are comparing it to nuclear DNA. After running gel electrophoresis, you notice that the mtDNA genome is smaller than the nuclear genome. You decide to examine the mtDNA and uncover that the mitochondrial genome contains genes coding for the production of rRNAs, ATP synthase, and citrate synthase. You identified a protein called *TFAM* (protein involved in mitochondrial genome replication) present in cells before running the gel, but you cannot seem to find the appropriate sequence in the mtDNA. What is the most likely reason you cannot find the TFAM gene?
    a) Mitochondrial proteins need to constantly be phosphorylated in order to maintain their proper structure.
    b) The TFAM gene from the mtDNA was translocated into the nuclear DNA of the cell.
    c) The mitochondrial genome is redundant with the nuclear genome and lost the TFAM gene as a result.
    d) The TFAM gene is located in the nuclear genome, and the proteins are transported to the mitochondria.
    e) TFAM is synthesized in the cytosol after the mitochondrial mRNA transcript leaves the organelle and then is returned back to the mitochondria once completed.

22) Geraniums are a unique plant species which follow a biparental mode of inheritance for chloroplasts. Your local botanist has decided to cross wild type green-leafed geraniums with mutant white-leafed geraniums. The resulting offspring include phenotypes of green-leafed plants, white-leafed plants, and variegated plants. What is the most likely explanation for the variegated geraniums?
    a) The geranium gametes mutated to produce the variegated offspring.
    b) The cpDNA recombined with the nuclear DNA of the plant cells.
    c) The parent generation were infected by a virus that contaminated the offspring.
    d) The variegated geraniums are a result of heteroplasmy due to the mode of inheritance.
    e) The chloroplast alleles from each parent are codominant with one another.

23) Your partner has decided to use a 2 color DNA microarray to test the water retention of a plant. They extract mRNA from a well-watered plant and drought-stressed plant of the same species, reverse transcribe it into cDNA using fluorescent labels, and then hybridizes the cDNA to a microarray slide. A red fluorescence indicates a drought-stressed plant, and a green fluorescence indicates a well-watered plant. When they go to examine the slide, they notice yellow fluorescence from some of the spots on the slide. What is the reason for the yellow spots?

a) RNA has degraded in the sample due to an error in preparation.
b) The microarrays were not properly labeled with the fluorescent markers.
c) Residual genomic DNA bound to the microarrays and are synthesizing an unintended marker.
d) There is an equal expression of well-watered and drought-stressed proteins.
e) Nonspecific hybridization occurred with the RNA.

24) Use the image for reference to the following scenario.



A classroom is doing a lab experiment where they are doing RFPL analysis of a tomato. They each extract the DNA from their tomato, digest it with EcoRI (a restriction enzyme), and run it on the agarose gel above. All but 1 student (the middle lane) observes the expected results. What is the most likely reason for the student's sample in the middle lane for showing a single band? Note: A "-" and "+" sign have been included for orientation of the gel.

a) EcoRI was unable to digest the DNA due to incomplete digestion.
b) The agarose concentration was too dense to allow for DNA fragments to properly move in the middle lane.
c) The ethidium bromide was unable to bind to the fragments in the middle lane.
d) The DNA sample had too many restriction enzymes added, fragmenting the DNA into undetectable pieces.
e) The tomato lacked any genetic variation for the restriction enzyme to properly cut.

Section 2: Free Response Questions

Answer the questions based on the information given. Points are earned both for a correct answer and explanation to support that answer. Point totals vary by question.

1) Before inserting recombinant plasmids into bacteria, it is important to check that your plasmids are the correct length. One way to do this is by running samples through a gel. You decide to perform a gel electrophoresis procedure on 5 samples from 5 containers of the same plasmid (1 individual sample per container in each lane). You ask your assistant to perform the procedure for you and come back to you with the results. When they return, they tell you that lanes 1, 3, 4, and 5 showed the expected results while lane 2 had a result indicating the plasmid had fewer base pairs. Much to your confusion, you perform Sanger sequencing on the sample from every lane and see the sequences in all lanes are identical. What is a possible reason lane 2's band does not match the expected results? Explain your answer.
   Note: Assume all lab procedures were done correctly and there was no contamination (or anything similar) throughout the process. (4)

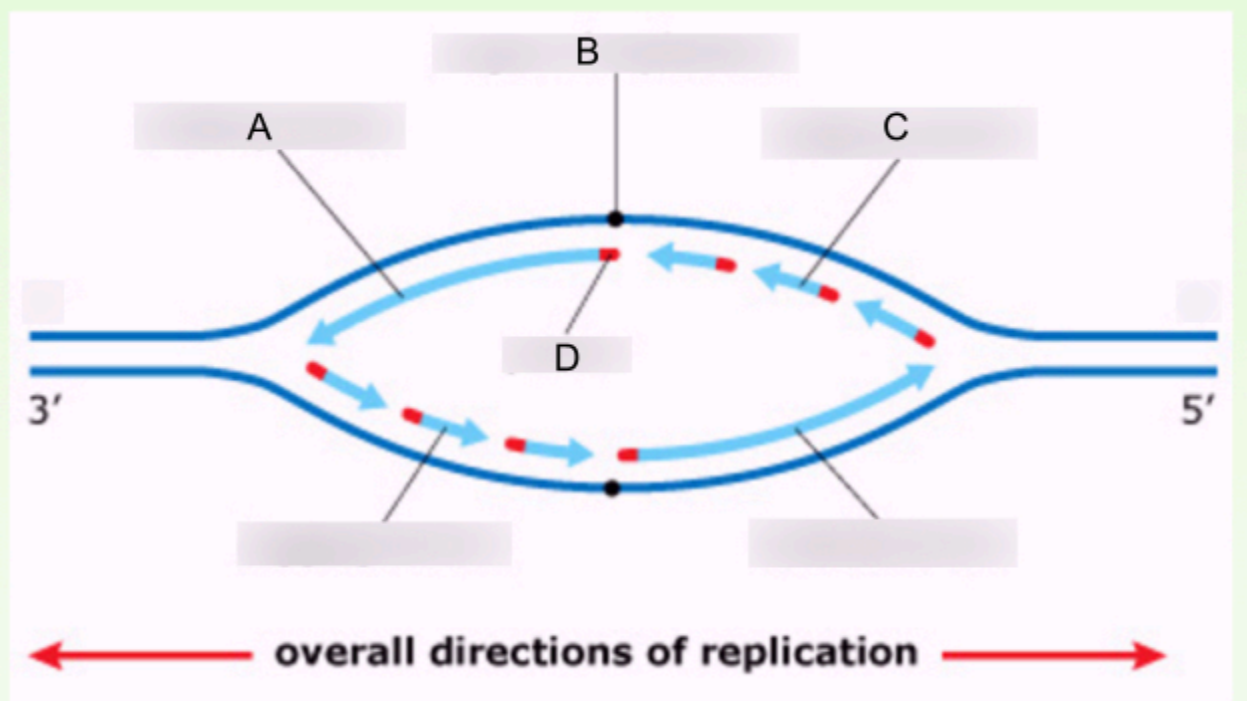
2) Why do females (XX chromosomes) have the same amount of X gene products as males (XY) despite having 2 times the number of X chromosomes? What is this process called? (3)

3) Use your knowledge from question #2 to answer this question: A female offspring is confirmed to have an X-linked disease at birth. After 20 years, cells are harvested from this individual. Doctors find that there are both healthy and unhealthy cells. Should this finding surprise them? Why or why not? (4)

4) A group of 40 people (20 male and 20 female) are stranded on a deserted island with no means to get help. The group decides to repopulate the island. Three of your friends are heterozygous for cystic fibrosis. Cystic fibrosis is an autosomal recessive disorder and will be noted as c. The frequency of this allele does not change as your population grows, what is the expected incidence of cystic fibrosis on your island? Show all your work and mention any relevant assumptions being made. Round your answer to 4 decimal places. (4)

5) Does acetylation increase or decrease gene expression? What specifically is getting acetylated? Describe how DNA is physically during the process of acetylation. (5)

6) Below is an image of a replication bubble. Complete the replication bubble (A-D) with the appropriate vocabulary. (4)



7) What are all the possible gamete genotypes of an animal with the genotype AaBb? Explain why there are multiple possibilities and why gametes have fewer alleles than their parents. (3)

Section 3: Design A Gene!

As an ~~evil~~ ethical scientist, you're going to help me (your lovely and awesome test writer) create a gene by answering some important questions! However, before we let you do that, I have to make sure you know what you're doing. Answer the following questions as you progress through our journey of making a gene.

1) Fill in the blanks (each blank is 1 point; 20 total blanks):
   The central dogma of biology involves flow of information from _____ to _____ to _____. The first step of this process is _____. This step starts when the enzyme _____ binds to a _____ region. Two important bacterial promoters are the _____ elements. An important human promoter is the _____. These regions contain DNA sequences which signal the enzyme to start the process. The next step is called _____, in which the enzyme moves in the _____ direction while adding to the mRNA strand in the _____ direction. The DNA strand that the enzyme moves down is called the _____ strand while the other is called the _____ strand. In humans, before the next process begins, the resulting strand from the first process is altered through 3 main processes: _____. The final step of this process is called _____ and occurs when the enzyme reaches a _____ sequence. The two main strategies for this in bacteria are _____ and _____. The next process in the central dogma is called _____. In bacteria, the resulting strand from the first process is read by _____.

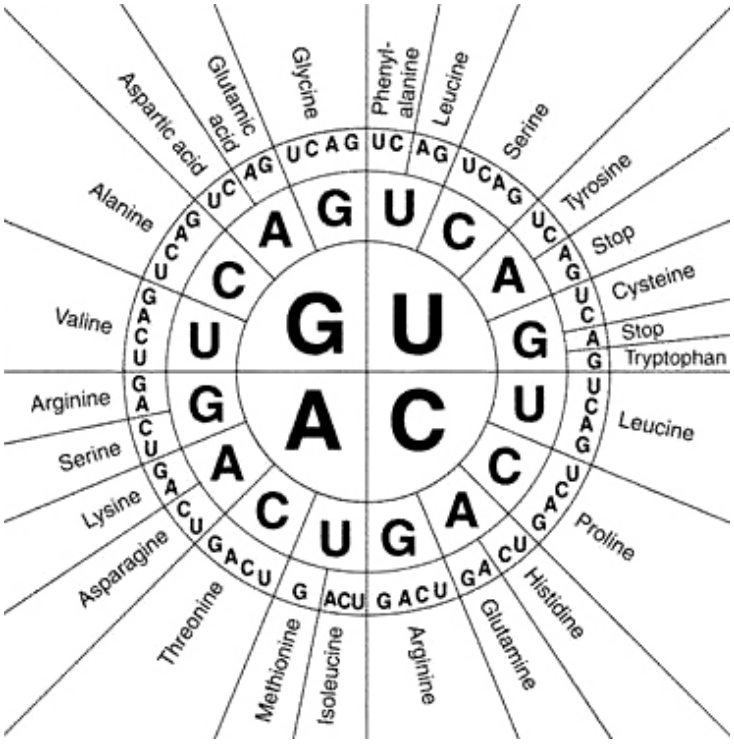
2) Great job so far! Let's get to the (hopefully) fun stuff. I want to make my genes into bacteria. That means it's going to be an operon. Unfortunately for you, I don't remember the order of sites of an operon. I'm planning on modeling mine after the lac operon, so if you could write down the order of relevant sites from left to right on the line below I'd be delighted! (5)

_____

3) The start of my gene is displayed in the image below. Armed with the knowledge you possess (and this handy dandy codon wheel), write out the correct amino acid sequence. The bottom strand is the coding strand (and make sure to include your primes and terminuses when writing your answer). Show all of your work from the image below to your final answer. Remember, this gene is going to be part of a bacteria! (8)

3' T A G C C C G T A T T C... 5'

5' A T C G G G C A T A A G... 3'



An image of the recombinant plasmid is shown below. The black region is the origin of replication (ori). The light blue region is an operon with my gene that contains a promoter sequence. The green region is a special promoter region that can only be activated when the light blue region is being transcribed. The red region is the gene for a fluorescent protein. The dark blue region is a gene encoding antibiotic resistance. Image credits: AMGEN.



4) I've been recommended by a colleague to test using this recombinant plasmid to test if my protein of interest is being produced. Why is this plasmid useful in that regard? (3)

5) Almost there! Making this plasmid wasn't cheap. It cost around $10,000,000 to produce a single one. Unfortunately, I don't have any more money to spend on making more. What is the most efficient way I can produce more of this plasmid? Detail the steps of the process (1 sentence per step with temperatures in celsius) (4)

6) There's a special enzyme used in the process you detailed in the last question. What is the name of that enzyme and why do we use it as opposed to something else? (2)

7) Final question! Alright, thanks to you I have a lot of this special plasmid. Your last question is to tell me whether or not I should put this in bacterial or eukaryotic cells? List 2 reasons. (2)